



DR ESLAM AHMED
eliteigcse.com

Kinematics Graphs

Sketch the graph first, label every number, then read it: the gradient is the acceleration and the area is the distance.

THE TOPIC ON ONE MAP

Kinematics Graphs

01 Gradient and area

gradient = a

area = distance

below the axis

02 One journey

trapezium

find T or V

quadratic in T

03 Two journeys, same axes

same speed

catches up

overtakes

04 Other graphs

$a-t$

$s-t$

impulse at a jump

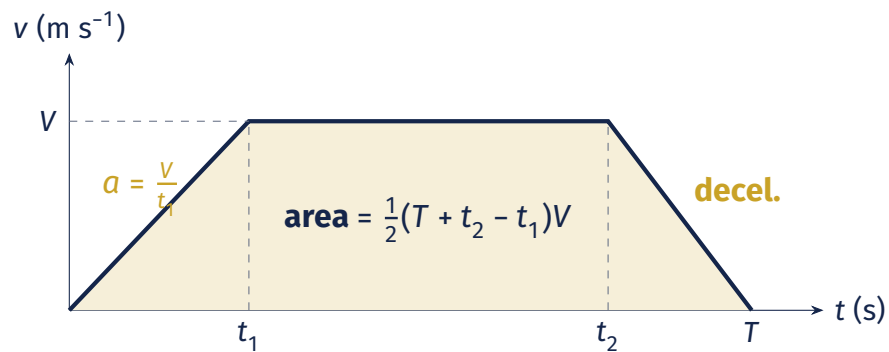
THE TWO FACTS OF EVERY SPEED-TIME GRAPH

gradient = acceleration

steeper line = bigger acceleration

area = distance

split into triangles, rectangles, trapezia



01 Gradient and Area

Every question on a speed–time or velocity–time graph uses one of two facts. Gradient = acceleration (negative gradient = deceleration). Area = distance; for velocity, area **below** the axis is motion backwards.

WHEN YOU SEE → YOU DO

“find the acceleration / deceleration”

→ gradient = $\frac{\text{rise}}{\text{run}}$; deceleration is the size

“find the distance”

→ area — split into simple shapes

the velocity line crosses the t -axis

→ at rest there; area below = moving backwards

“thrown vertically upwards ... velocity–time graph”

→ gradient = -9.8 ; area above the axis = height gained

“total distance” vs “displacement”

→ add all areas vs areas with signs

WORKED MODEL • OCTOBER 2021

A ball is thrown upwards from A; its velocity reaches 0 at $t = 2.5$ and it hits the ground at $t = 7$. Find the launch speed and the height of A.

1 $u = 9.8 \times 2.5 = 24.5 \text{ m s}^{-1}$

gradient = $-g$

2 at $t = 7$: $v = 24.5 - 9.8(7) = -44.1$

velocity line

ANSWER $\frac{1}{2}(4.5)(44.1) - \frac{1}{2}(2.5)(24.5) = 99.2 - 30.6 = 68.6 \text{ m}$

IDEA 01

Practise: Gradient and Area

My questions first, then real past papers.

Q1 DR ESLAM Gradient and area

M1 · TOPIC 3 3 marks

A particle speeds up uniformly from 4 m s^{-1} to 10 m s^{-1} in 3 s, then moves at 10 m s^{-1} for 5 s. Find its acceleration and the total distance.

Gradient = acceleration; area = distance (trapezium + rectangle).

Q2 DR ESLAM Below the axis

M1 · TOPIC 3 4 marks

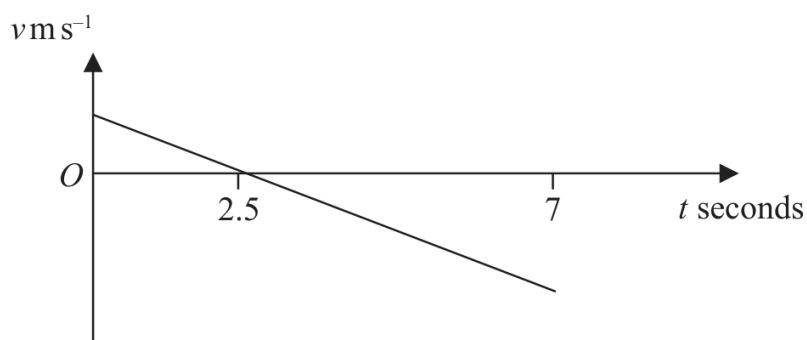
The velocity of a particle changes at a constant rate from 12 m s^{-1} to -6 m s^{-1} in 9 s. Find when it is at rest, its displacement and the distance it travels.

PAST PAPERS

Q3 PAST PAPER Pearson IAL WME01 • October 2021 • Q6

M1 • TOPIC 3 5 marks

6.

**Figure 3**

A small ball is thrown vertically upwards at time $t = 0$ from a point A which is above horizontal ground. The ball hits the ground 7 s later.

The ball is modelled as a particle moving freely under gravity.

The velocity-time graph shown in Figure 3 represents the motion of the ball for $0 \leq t \leq 7$

(a) Find the speed with which the ball is thrown.

(2)

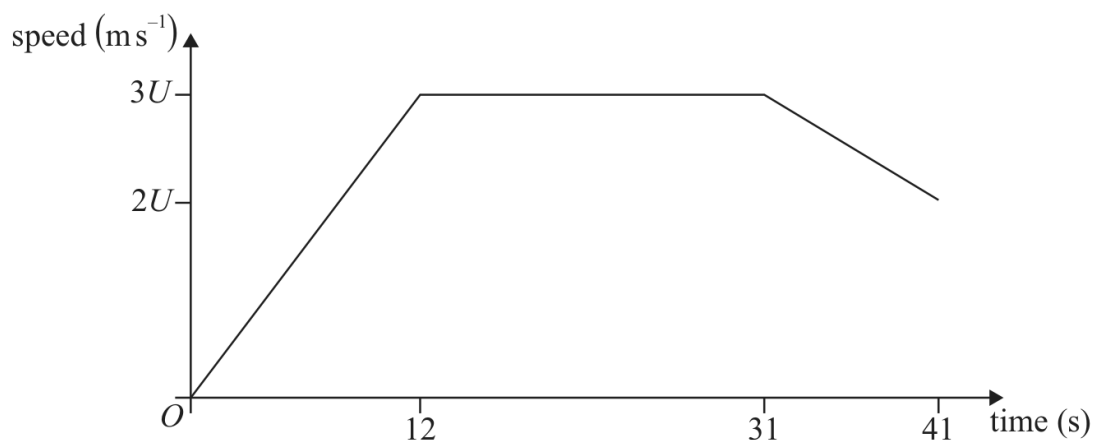
(b) Find the height of A above the ground.

(3)

Q4 PAST PAPER Pearson IAL WME01 • January 2026 • Q2

M1 • TOPIC 3 10 marks

2.

**Figure 1**

A particle P starts from rest at time $t = 0$ and moves in a straight line

- from $t = 0$ to $t = 12$ s, P accelerates uniformly until it reaches a speed of $3U \text{ m s}^{-1}$
- from $t = 12$ s to $t = 31$ s, P moves with constant speed $3U \text{ m s}^{-1}$
- from $t = 31$ s to $t = 41$ s, P decelerates uniformly until it has speed $2U \text{ m s}^{-1}$

as shown on the speed-time graph in Figure 1.

The distance travelled by P between $t = 0$ and $t = 41$ s is 600 m.

- Show that $U = 6$ (3)
- Find the distance travelled by P whilst moving from rest to a speed of 6 m s^{-1} (2)
- Find the acceleration of P between $t = 0$ and $t = 12$ s. (1)
- Find the deceleration of P between $t = 31$ s and $t = 41$ s. (2)
- Sketch an acceleration-time graph to represent the motion of P from $t = 0$ to $t = 41$ s. (2)

02 One Journey: Sketch, Then Use the Area

Draw the journey as straight lines: speeding up, steady, slowing down. Put every number from the question on the graph. Then one equation from the area (the distance) and one from the times.

WHEN YOU SEE → YOU DO

“sketch a speed–time graph”

→ straight segments, label every speed and time

“the total distance is 4.8 km”

→ area = 4800 — change units first

“the deceleration is twice the acceleration”

→ deceleration time = half the acceleration time

“show that $T^2 - kT + 500 = 0$ ”

→ area in T , then tidy into the quadratic

“hence find T , justifying”

→ reject the root that breaks the story ($T > \text{total time}$)

Sketch and label

Area = distance

Solve for T or V

Check the story

WORKED MODEL • JANUARY 2023

From rest to 20 m s^{-1} in $T \text{ s}$, 20 m s^{-1} for 180 s , then deceleration twice the acceleration. Distance 4.8 km . Find T .

1 slowing down takes $\frac{T}{2}$

twice the rate, half the time

2 $\frac{1}{2}(20)T + 20(180) + \frac{1}{2}(20)\frac{T}{2} = 4800$

area

3 $15T = 1200$

collect

ANSWER $T = 40 \text{ s}$

× **TRAP** Forgetting to change 4.8 km into 4800 m , or minutes into seconds, before writing the area.

IDEA 02

Practise: One Journey: Sketch, Then Use the Area

My questions first, then real past papers.

Q5 DR ESLAM Trapezium: find V

M1 • TOPIC 3 3 marks

A train accelerates from rest to $V \text{ m s}^{-1}$ in 20 s, travels at V for 60 s, then decelerates to rest in 10 s. It travels 1.35 km. Find V .

Trapezium area = $\frac{1}{2}(\text{top} + \text{bottom}) \times V$.

$V =$ _____

Q6 DR ESLAM Times from accelerations

M1 • TOPIC 3 4 marks

A car accelerates from rest at 2 m s^{-2} to 20 m s^{-1} , keeps this speed, then decelerates at 4 m s^{-2} to rest. The whole journey takes 60 s. Find the distance.

PAST PAPERS

Q7 PAST PAPER Pearson IAL WME01 · October 2019 · Q6

M1 · TOPIC 3 12 marks

6. An athlete runs a 200 m race along a straight horizontal track.

In a model of the motion of the athlete, air resistance is ignored, the athlete starts from rest at time $t = 0$ seconds and moves with uniform acceleration 0.8 m s^{-2} for T seconds, reaching a speed of $V \text{ m s}^{-1}$. She then maintains this speed until she crosses the finishing line.

The total time from when the athlete starts to when she crosses the finishing line is 30 s.

- (a) Sketch a speed-time graph for the model of the motion of the athlete from the instant when she starts to the instant when she crosses the finishing line. (2)
- (b) Write down an expression for V in terms of T . (1)
- (c) Show that $T^2 - kT + 500 = 0$, where k is a constant to be found. (4)
- (d) Hence find the value of T , justifying your answer carefully. (3)
- (e) Considering your speed-time graph or otherwise, state two ways, apart from including air resistance, in which the model could be made to be more realistic. (2)

Q8 PAST PAPER Pearson IAL WME01 • January 2021 • Q7

M1 • TOPIC 3 12 marks

7. A helicopter is hovering at rest above horizontal ground at the point H . A parachutist steps out of the helicopter and immediately falls vertically and freely under gravity from rest for 2.5 s. His parachute then opens and causes him to immediately decelerate at a constant rate of 3.9 m s^{-2} for T seconds ($T < 6$), until his speed is reduced to $V \text{ m s}^{-1}$. He then moves with this constant speed $V \text{ m s}^{-1}$ until he hits the ground. While he is decelerating, he falls a distance of 73.75 m. The total time between the instant when he leaves H and the instant when he hits the ground is 20 s.

The parachutist is modelled as a particle.

- (a) Find the speed of the parachutist at the instant when his parachute opens. (1)
- (b) Sketch a speed-time graph for the motion of the parachutist from the instant when he leaves H to the instant when he hits the ground. (2)
- (c) Find the value of T . (5)
- (d) Find, to the nearest metre, the height of the point H above the ground. (4)

Q9**PAST PAPER**

Pearson IAL WME01 • January 2023 • Q1

M1 • TOPIC 3**10 marks**

1. A train travels along a straight horizontal track between two stations A and B .

The train starts from rest at station A and accelerates uniformly for T seconds until it reaches a speed of 20 m s^{-1}

The train then travels at a constant speed of 20 m s^{-1} for 3 minutes before decelerating uniformly until it comes to rest at station B .

The magnitude of the acceleration of the train is twice the magnitude of the deceleration.

- (a) On the axes below, sketch a speed–time graph to illustrate the motion of the train as it moves from station A to station B .



If you need to redraw your graph, use the axes on page 3

(3)

Stations A and B are 4.8 km apart.

- (b) Find the value of T

(5)

- (c) Find the acceleration of the train during the first T seconds of its motion.

(2)

Only use these axes if you need to redraw your graph.



(Total for Question 1 is 10 marks)

Q10**PAST PAPER**

Pearson IAL WME01 · January 2025 · Q2

M1 · TOPIC 3 11 marks

2. The fixed points A , B and C lie in a straight line on a horizontal road.

- At time $t = 0$, a motorbike passes through A with speed 5 m s^{-1}
- From A , the motorbike accelerates uniformly until it reaches B with a speed of $V \text{ m s}^{-1}$
- The motorbike takes T_1 seconds to travel from A to B
- From B , the motorbike decelerates uniformly until it comes to rest at C
- The motorbike takes T_2 seconds to travel from B to C

(a) Sketch a speed-time graph for the motion of the motorbike as it moves from A to C .

(3)

The distance AB is 132 m and the distance BC is 136 m.

(b) Find, in terms of V , an expression for

(i) T_1

(ii) T_2

(4)

Given that the motorbike takes 28 s to travel from A to C ,

(c) find the value of V ,

(2)

(d) find the deceleration of the motorbike.

(2)

03 Two Journeys on the Same Axes

Two vehicles on one graph. **Same speed** is where the two lines cross. **Same distance** (catches up, overtakes) is where the two **areas** are equal — not where the lines cross.

WHEN YOU SEE → YOU DO

“sketch on the same axes”

→ both graphs, one diagram, key points labelled

“at the instant they are moving with the same speed”

→ the lines cross: equal v

“catches up with” / “overtakes” / “passes”

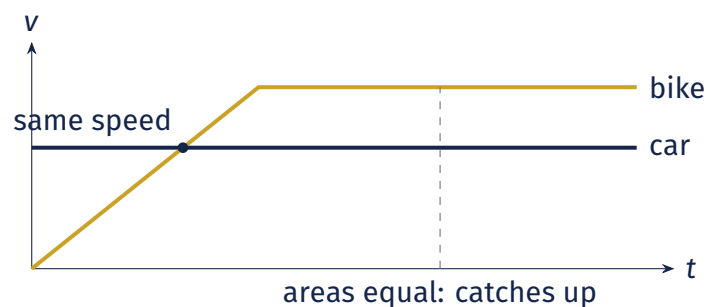
→ equal areas from the start

“starts 20 seconds later”

→ its graph starts at $t = 20$ on the axis

“how far in front is A?”

→ difference of the two areas



WORKED MODEL • JANUARY 2020

A car at a constant 40 m s^{-1} passes a motorcyclist at rest. She accelerates at 4 m s^{-2} for 15 s. When are their speeds equal?

1 motorcyclist: $v = 4t$

gradient 4

ANSWER $4t_1 = 40 \Rightarrow t_1 = 10 \text{ s}$

× **TRAP** Answering “catches up” with the time the lines cross. That is only when the speeds are equal.

IDEA 03

Practise: Two Journeys on the Same Axes

My questions first, then real past papers.

Q11 **DR ESLAM** Same speed, then catch up

M1 · TOPIC 3 **5 marks**

At $t = 0$ a car passes a motorcycle at rest, moving at a constant 15 m s^{-1} . The motorcycle accelerates at 3 m s^{-2} to 24 m s^{-1} , then keeps that speed. Find (a) when they have the same speed (b) when the motorcycle catches the car. Same speed: where the lines cross. Catch up: where the AREAS are equal.

PAST PAPERS

Q12 PAST PAPER Pearson IAL WME01 · January 2020 · Q5

M1 · TOPIC 3 10 marks

5. A car travels at a constant speed of 40 ms^{-1} in a straight line along a horizontal racetrack. At time $t = 0$, the car passes a motorcyclist who is at rest. The motorcyclist immediately sets off to catch up with the car.

The motorcyclist accelerates at 4 ms^{-2} for 15 s and then accelerates at 1 ms^{-2} for a further T seconds until he catches up with the car.

- (a) Sketch, on the same axes, the speed-time graph for the motion of the car and the speed-time graph for the motion of the motorcyclist, from time $t = 0$ to the instant when the motorcyclist catches up with the car.

(2)

At the instant when $t = t_1$ seconds, the car and the motorcyclist are moving at the same speed.

- (b) Find the value of t_1

(2)

- (c) Show that $T^2 + kT - 300 = 0$, where k is a constant to be found.

(6)

Q13**PAST PAPER**

Pearson IAL WME01 • June 2021 • Q8

M1 • TOPIC 3 13 marks

8. Two trams, tram A and tram B , run on parallel straight horizontal tracks. Initially the two trams are at rest in the depot and level with each other.

At time $t = 0$, tram A starts to move. Tram A moves with constant acceleration 2 m s^{-2} for 5 seconds and then continues to move along the track at constant speed.

At time $t = 20$ seconds, tram B starts from rest and moves in the same direction as tram A . Tram B moves with constant acceleration 3 m s^{-2} for 4 seconds and then continues to move along the track at constant speed.

The trams are modelled as particles.

- (a) Sketch, on the same axes, a speed-time graph for the motion of tram A and a speed-time graph for the motion of tram B , from $t = 0$ to the instant when tram B overtakes tram A .

(3)

At the instant when the two trams are moving with the same speed, tram A is d metres in front of tram B .

- (b) Find the value of d .

(5)

- (c) Find the distance of the trams from the depot at the instant when tram B overtakes tram A .

(5)**(Total 13 marks)**

Q14 PAST PAPER Pearson IAL WME01 · June 2022 · Q7

M1 · TOPIC 3 13 marks

7. Two small children, Ajaz and Beth, are running a 100 m race along a straight horizontal track.

They both start from rest, leaving the start line at the same time.

Ajaz accelerates at 0.8 m s^{-2} up to a speed of 4 m s^{-1} and then maintains this speed until he crosses the finish line.

Beth accelerates at 1 m s^{-2} for T seconds and then maintains a constant speed until she crosses the finish line.

Ajaz and Beth cross the finish line at the same time.

- (a) Sketch, on the same axes, a speed-time graph for each child, from the instant when they leave the start line to the instant when they cross the finish line. (3)

- (b) Find the time taken by Ajaz to complete the race. (4)

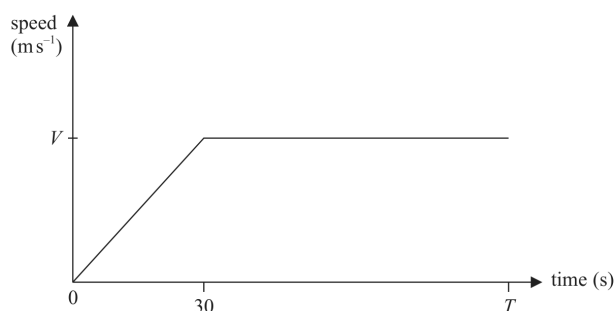
- (c) Find the value of T (4)

- (d) Find the difference in the speeds of the two children as they cross the finish line. (2)

Q15 PAST PAPER Pearson IAL WME01 • June 2023 • Q5

M1 • TOPIC 3 11 marks

5.

**Figure 2**

The speed-time graph in Figure 2 illustrates the motion of a car travelling along a straight horizontal road.

At time $t = 0$, the car starts from rest and accelerates uniformly for 30 s until it reaches a speed of $V \text{ m s}^{-1}$.

The car then travels at a constant speed of $V \text{ m s}^{-1}$ until time $t = T$ seconds.

- (a) Show that the distance travelled by the car between $t = 0$ and $t = T$ seconds is $V(T - 15)$ metres.

(2)

A motorbike also travels along the same road.

- The motorbike starts from rest at time $t = 10 \text{ s}$ and accelerates uniformly for 40 s
- The acceleration of the motorbike is the **same** as the acceleration of the car
- The motorbike then travels at a constant speed for a further 10 s before decelerating uniformly until it reaches a speed of $V \text{ m s}^{-1}$ at time T seconds

- (b) On Figure 2, sketch a speed-time graph for the motion of the motorbike.

[If you need to redraw your sketch, there is a copy of Figure 2 on page 15.]

(2)

- (c) Show that the constant speed of the motorbike is $\frac{4V}{3} \text{ m s}^{-1}$

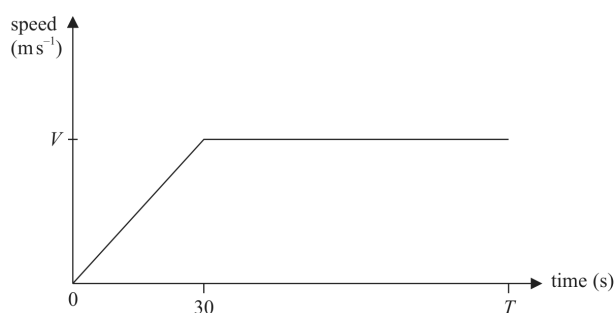
(2)

At time $t = T$ seconds, the distance travelled by each vehicle is the same.

- (d) Find the value of T

(5)

Only use this copy of Figure 2 if you need to redraw your sketch.

**Figure 2**

(Total for Question 5 is 11 marks)

Q16 PAST PAPER Pearson IAL WME01 • October 2024 • Q4

M1 • TOPIC 3 13 marks

4. The points A and B lie on the same straight horizontal road.

Figure 2, on page 11, shows the speed-time graph of a cyclist P , for his journey from A to B .

At time $t = 0$, P starts from rest at A and accelerates uniformly for 9 seconds until his speed is $V \text{ m s}^{-1}$.

He then travels at constant speed $V \text{ m s}^{-1}$.

When $t = 42$, cyclist P passes B .

Given that the distance AB is 120 m,

- (a) show that $V = 3.2$ (3)

- (b) Find the acceleration of cyclist P between $t = 0$ and $t = 9$ (2)

Cyclist P continues to cycle along the road in the same direction at the same constant speed, $V \text{ m s}^{-1}$.

When $t = 6$, a second cyclist Q sets off from A and travels in the same direction as P along the same road. She accelerates for T seconds until her speed is 3.6 m s^{-1} .

She then travels at constant speed 3.6 m s^{-1} .

Cyclist Q catches up with P when $t = 54$.

- (c) On Figure 2, on page 11, sketch a speed-time graph showing the journeys of both cyclists, for the interval $0 \leq t \leq 54$ (3)

- (d) Find the value of T (5)

Question 4 continued

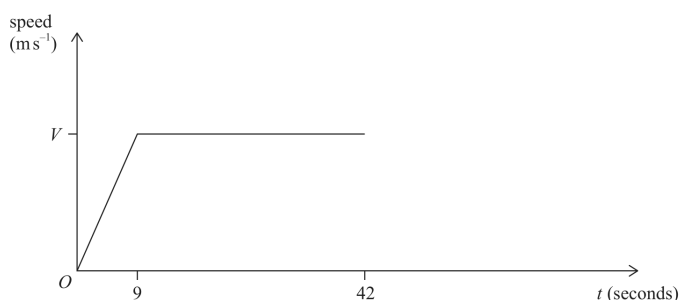
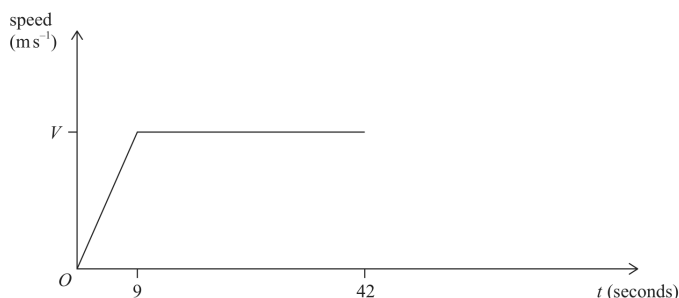


Figure 2

A copy of Figure 2 is on page 13 if you need to redraw your answer to part (c).

Question 4 continued

Only use this copy of Figure 2 if you need to redraw your answer to part (c).



Copy of Figure 2

04 Acceleration–Time and Displacement–Time Graphs

Change the graph back into a speed–time graph first. On an acceleration–time graph the **area** is the change in speed. On a displacement–time graph the **gradient** is the velocity.

WHEN YOU SEE → YOU DO

an acceleration–time graph

→ area = change of speed; build v step by step

“sketch an acceleration–time graph”

→ horizontal steps: gradient of each v – t part

a displacement–time graph

→ gradient = velocity on each straight part

“impulse at $t = 0.3$ ” on an s – t graph

→ $m(v_{\text{after}} - v_{\text{before}})$

graph	gradient means	area means
displacement–time	velocity	—
velocity–time	acceleration	displacement
acceleration–time	—	change of velocity

WORKED MODEL • OCTOBER 2023

$v = 8$ at $t = 0$; $a = -0.5$ for 4 s, $a = 0$ for 6 s, $a = 1.2$ for 10 s. Show $v = 18$ at $t = 20$.

1 $8 - 0.5(4) = 6$

area of the first step

2 $6 + 0 + 1.2(10) = 18$

add the other areas

ANSWER $v = 18 \text{ m s}^{-1}$

IDEA 04

Practise: Acceleration–Time and Displacement–Time Graphs

My questions first, then real past papers.

Q17 **DR ESLAM** From an acceleration–time graph

M1 • TOPIC 3 4 marks

Starting from rest, $a = 2 \text{ m s}^{-2}$ for $0 \leq t \leq 5$, $a = 0$ for $5 \leq t \leq 15$, $a = -1 \text{ m s}^{-2}$ for $15 \leq t \leq 20$. Find the speed at $t = 20$ and the distance travelled.

Area under the a – t graph = change of speed.

Q18 **DR ESLAM** From a displacement–time graph

M1 • TOPIC 3 3 marks

A displacement–time graph is made of straight lines through $(0, 0)$, $(4, 12)$, $(10, 12)$ and $(14, 0)$. Find the velocity on each part, the total distance and the final displacement.

PAST PAPERS

Q19 PAST PAPER Pearson IAL WME01 • October 2023 • Q2

M1 • TOPIC 3 10 marks

2.

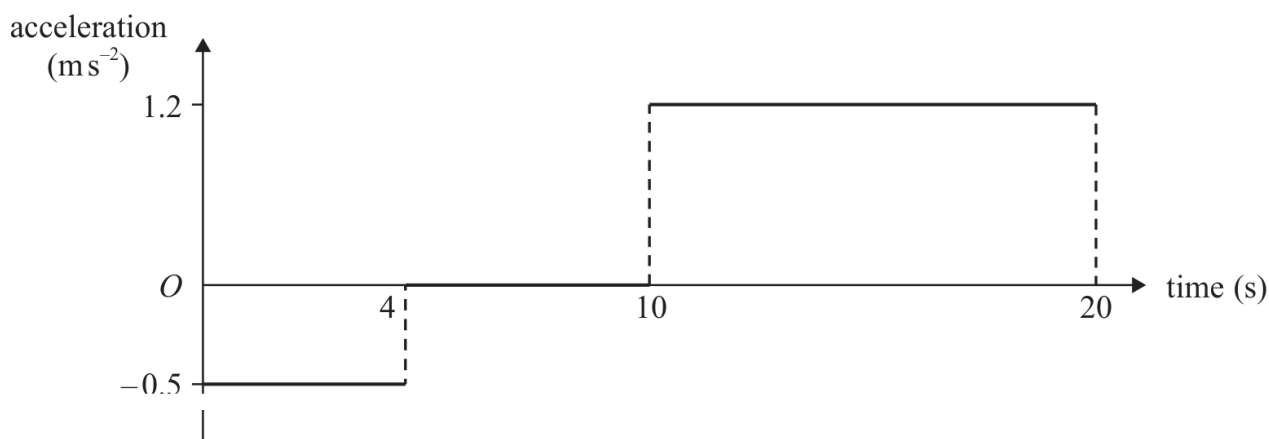


Figure 2

Two fixed points, A and B , are on a straight horizontal road.

The **acceleration-time** graph in Figure 2 represents the motion of a car travelling along the road as it moves from A to B .

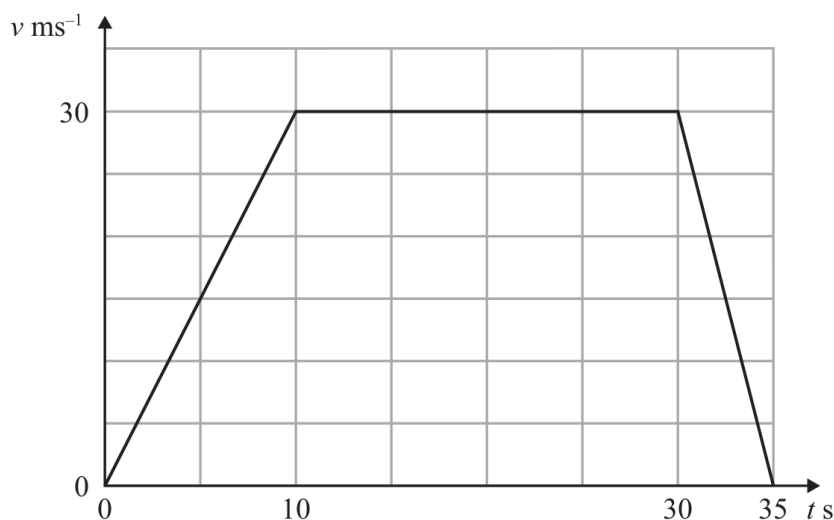
At time $t = 0$, the car passes through A with speed 8 m s^{-1}

At time $t = 20 \text{ s}$, the car passes through B with speed $v \text{ m s}^{-1}$

- (a) Show that $v = 18$ (3)
- (b) Sketch a speed-time graph for the motion of the car from A to B . (3)
- (c) Find the distance AB . (4)

Q20 PAST PAPER Pearson IAL WME01 • October 2025 • Q6**M1 • TOPIC 3 10 marks**

6.

**Figure 4**

The point O lies on a straight horizontal road.
At time $t = 0$, a car leaves O and travels along the road.

The velocity-time graph in Figure 4 shows the velocity, $v \text{ ms}^{-1}$, of the car at time t seconds for the first 35 seconds of its journey.

(a) Find

(i) the acceleration of the car for the period $0 \leq t \leq 10$ (ii) the deceleration of the car for the period $30 \leq t \leq 35$ **(2)**(b) Sketch an acceleration-time graph for the car for the period $0 \leq t \leq 35$ **(2)**(c) Find the distance travelled by the car for the period $0 \leq t \leq 35$ **(2)**

When $t = 5$, a motorcycle starts from rest at O .

The motorcycle travels along the same road as the car and in the same direction.

For the period $5 \leq t \leq 20$, the acceleration of the motorcycle is $A \text{ ms}^{-2}$, where A is a positive constant.

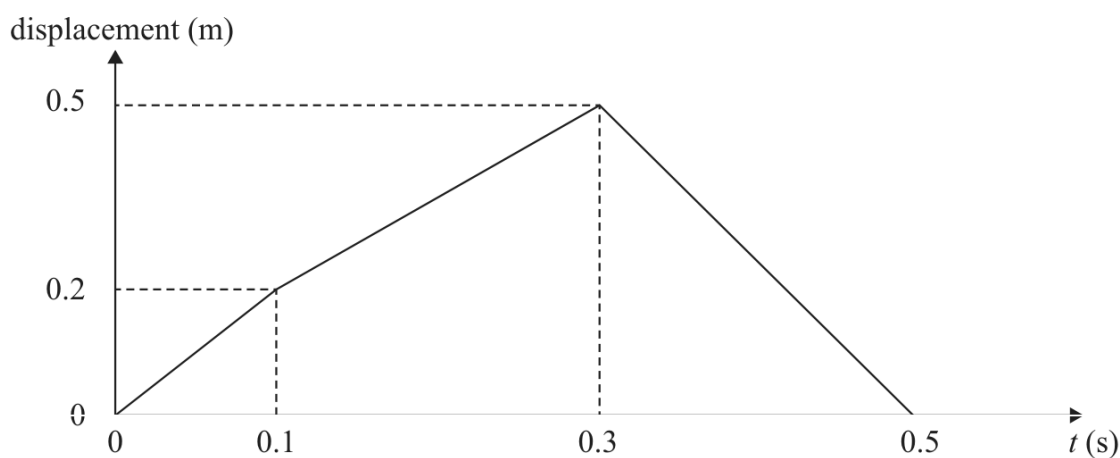
The motorcycle catches up with the car when $t = 20$

(d) Find the value of A .**(4)**

Q21 PAST PAPER Pearson IAL WME01 • June 2026 • Q3

M1 • TOPIC 3 10 marks

3.

**Figure 2**

A small ball of mass 0.04 kg moves along a straight line on a pinball table.

At time $t = 0$, the ball passes through the point O

The displacement of the ball is measured from O

Figure 2 shows the **displacement-time** graph for the motion of the ball from $t = 0$ to $t = 0.5$

- (a) State the displacement of the ball from O at $t = 0.5$ (1)
- (b) Find the total distance moved by the ball between $t = 0$ and $t = 0.5$ (1)
- (c) Sketch a velocity-time graph for the motion of the ball from $t = 0$ to $t = 0.5$ (4)
- (d) Find the magnitude of the impulse exerted on the ball at $t = 0.3$, stating the units of your answer. (4)

(Total for Question 3 is 10 marks)

ALL The Whole Topic on One Page

Tear this page out before the exam.

- | | | |
|-----------|--------------------------|--|
| 01 | Gradient and area | Gradient = a ; area = distance; below the axis = backwards. |
| 02 | One journey | Sketch and label; area = distance; reject the impossible root. |
| 03 | Two journeys | Same speed: lines cross. Catch up: areas equal. |
| 04 | Other graphs | $a-t$ area = Δv ; $s-t$ gradient = v . |

BOTTOM LINE Sketch → label → gradient for a , area for distance. Catch up = equal areas.

Answers

Kinematics Graphs • for the teacher. Every answer checked with a computer algebra system.

IDEA 01 • GRADIENT AND AREA

- Q1** Dr Eslam • Gradient and area $a = 2 \text{ m s}^{-2}$; $21 + 50 = 71 \text{ m}$
- Q2** Dr Eslam • Below the axis at rest at $t = 6$; displacement $36 - 9 = 27 \text{ m}$; distance $36 + 9 = 45 \text{ m}$
- Q3** Pearson IAL WME01 • October 2021 • Q6 (a) 24.5 m s^{-1} (b) 68.6 m
- Q4** Pearson IAL WME01 • January 2026 • Q2 (a) $U = 6$ (b) 12 m (c) 1.5 m s^{-2} (d) 0.6 m s^{-2} (e) steps 1.5 on 0–12, 0 on 12–31, -0.6 on 31–41

IDEA 02 • ONE JOURNEY: SKETCH, THEN USE THE AREA

- Q5** Dr Eslam • Trapezium: find V $\frac{1}{2}(90 + 60)V = 1350 \Rightarrow V = 18$
- Q6** Dr Eslam • Times from accelerations 10 s and 5 s, so 45 s at 20: $\frac{1}{2}(60 + 45)(20) = 1050 \text{ m}$
- Q7** Pearson IAL WME01 • October 2019 • Q6 (a) straight line to (T, V) , then level to $t = 30$ (b) $V = 0.8T$ (c) $k = 60$ (d) $T = 10$ ($T = 50 > 30$ rejected) (e) e.g. reaction time; acceleration not constant / speed not constant at the end
- Q8** Pearson IAL WME01 • January 2021 • Q7 (a) 24.5 m s^{-1} (b) three stages: up to 24.5, down to V , level (c) $T = 5$ (d) 167 m
- Q9** Pearson IAL WME01 • January 2023 • Q1 (a) $(0, 0) \rightarrow (T, 20) \rightarrow (T + 180, 20) \rightarrow (3T + 180, 0)$ (b) $T = 40$ (c) 0.5 m s^{-2}
- Q10** Pearson IAL WME01 • January 2025 • Q2 (a) $(0, 5) \rightarrow (T_1, V) \rightarrow (T_1 + T_2, 0)$ (b)(i) $T_1 = \frac{264}{V + 5}$ (ii) $T_2 = \frac{272}{V}$ (c) $V = 17$ (d) $\frac{17}{16} = 1.0625 \text{ m s}^{-2}$

IDEA 03 • TWO JOURNEYS ON THE SAME AXES

- Q11** Dr Eslam • Same speed, then catch up
(a) $t = 5 \text{ s}$ (b) $96 + 24(t - 8) = 15t \Rightarrow t = \frac{32}{3} \approx 10.7 \text{ s}$
- Q12** Pearson IAL WME01 • January 2020 • Q5
(a) car level at 40; motorcycle up to 60 at $t = 15$, then less steep to $15 + T$ (b) $t_1 = 10$ (c) $T^2 + 40T - 300 = 0$, $k = 40$
- Q13** Pearson IAL WME01 • June 2021 • Q8
(a) A: $(0, 0) \rightarrow (5, 10)$ then level; B: rest to $t = 20$, $(24, 12)$ then level (b) $d = \frac{575}{3} \approx 192 \text{ m}$ (c) 1170 m
- Q14** Pearson IAL WME01 • June 2022 • Q7
(a) Ajaz: up to 4 at $t = 5$, level; Beth: up to T at $t = T$, level (b) 27.5 s (c) $T = 3.92$ (d) 0.085 m s^{-1}
- Q15** Pearson IAL WME01 • June 2023 • Q5
(a) $\frac{1}{2}(30)V + V(T - 30) = V(T - 15)$ (b) through $(10, 0)$, $(50, \frac{4}{3}V)$, $(60, \frac{4}{3}V)$, (T, V) (c) $\frac{4V}{3}$ (d) $T = 90$
- Q16** Pearson IAL WME01 • October 2024 • Q4
(a) $V = 3.2$ (b) $\frac{16}{45} = 0.356 \text{ m s}^{-2}$ (c) P level to $(54, 3.2)$; Q through $(6, 0)$, $(6 + T, 3.6)$, $(54, 3.6)$ (d) $T = 8$

IDEA 04 • ACCELERATION–TIME AND DISPLACEMENT–TIME GRAPHS

- Q17** Dr Eslam • From an acceleration–time graph
 $v(20) = 5 \text{ m s}^{-1}$; $25 + 100 + 37.5 = 162.5 \text{ m}$
- Q18** Dr Eslam • From a displacement–time graph
 $3, 0, -3 \text{ m s}^{-1}$; distance 24 m ; displacement 0
- Q19** Pearson IAL WME01 • October 2023 • Q2
(a) $8 - 0.5(4) + 1.2(10) = 18$ (b) $(0, 8) \rightarrow (4, 6) \rightarrow (10, 6) \rightarrow (20, 18)$ (c) $AB = 184 \text{ m}$
- Q20** Pearson IAL WME01 • October 2025 • Q6
(a)(i) 3 m s^{-2} (ii) 6 m s^{-2} (b) steps $3, 0, -6$ with breaks at $10, 30, 35$ (c) 825 m (d) $A = 4$
- Q21** Pearson IAL WME01 • June 2026 • Q3
(a) 0 (b) 1 m (c) $v = 2$ on $0-0.1$, 1.5 on $0.1-0.3$, -2.5 on $0.3-0.5$ (d) $0.04(1.5 + 2.5) = 0.16 \text{ N s}$